

Lecture 20: Making Statistical Claims

*Instructor: Jonathan Pipping-Gamón**Authors: JP, RB*

20.1 From Statistical Models to Applications

Most of the course has focused on building statistical models. We estimated expected points, win probabilities, player abilities, team strengths, treatment effects, and player impact coefficients. Once those models are trained, one step remains: deciding what conclusions the findings justify.

That step is easy to skip. A model output can quickly turn into a claim like

- “The model says go for it on 4th down.”
- “This player is the eighth best player in the league.”
- “The timeout caused the comeback.”
- “This batting average proves the player improved.”
- “The neural network found what coaches should call.”

Each of these takeaways may come from disciplined analysis, but they may also overstate their conclusions by hiding the target, the uncertainty, and the assumptions that produced the estimate.

Claims should match the target, the model, and the uncertainty. This lecture focuses on two common places where sports models get overstated: fourth down recommendations and rankings built from estimated metrics.

20.2 Case Study 1: Fourth Down Decisions

Consider the fourth down decision problem. On fourth down, the offense can keep its offense on the field, attempt a field goal, or punt:

$$d \in \{\text{go}, \text{field goal}, \text{punt}\}.$$

Going for it can extend the drive, but a failed attempt gives the opponent the ball near the current spot. A field goal attempt can produce three points, but a miss gives up field position. A punt gives up the ball on purpose in exchange for forcing the opponent to start farther away from the end zone.

A model cannot recommend an action until we say what the action should optimize. A natural target is win probability. If x is the current game state, including score, time, field position, yards to go, and team information, the three decision values are

$$WP_{\text{go}}(x), \quad WP_{\text{FG}}(x), \quad WP_{\text{punt}}(x).$$

The counterfactual meaning of “would” matters here. In the actual game, the coach chooses one action. The other two outcomes are counterfactual. The model estimates what the team’s win probability would be after each possible action:

$$\widehat{WP}_{\text{go}}(x), \quad \widehat{WP}_{\text{FG}}(x), \quad \widehat{WP}_{\text{punt}}(x).$$

Under a win probability objective, the plug in rule is

$$\hat{d}(x) = \arg \max_{d \in \{\text{go}, \text{FG}, \text{punt}\}} \widehat{WP}_d(x).$$

The estimated edge for the selected action is

$$\hat{\Delta}(x) = \widehat{WP}_{\hat{d}}(x) - \max_{d \neq \hat{d}} \widehat{WP}_d(x).$$

This separates two things that are often mixed together. The model estimates decision values. The objective turns those values into a recommendation. That recommendation optimizes estimated win probability under the fitted model, not job security, injury risk, locker room trust, or future play calling value.

Tied, 4th & 3, 35 yards from opponent endzone								
Qtr 1, 10:28 Timeouts: Off 3, Def 3 Point Spread: -2.5, OQ-DQ: -0.49, KQ: 0, PQ: 0								
decision	WP	90% WP gain CI	boot %	success prob	WP if fail	WP if succeed	SD of WP	baseline coach %
Go for it	55.9%	[-1%, 5%]	89%	48%	47.9%	64.8%	8.4%	50.9%
Field goal	54.1%		11%	61%	47.0%	58.5%	5.6%	45.5%
Punt	52.4%		0%					3.6%

Figure 20.1: Example output from Ryan Brill's fourth down decision model for the Eagles' fourth-and-3 against the 49ers in the 2022 NFC Championship Game. The table compares the model's estimated win probability for each action.

20.2.1 Stability of the Recommendation

A fourth down recommendation usually combines several fitted pieces:

- conversion probability from past fourth down attempts;
- field goal make probability from distance and kicking context;
- punt value from field position and future scoring;
- win probability from score, clock, field position, teams, and remaining possessions.

Each piece includes estimation error. The data also omit football context that can matter: exact play call, injuries, fatigue, weather, personnel groupings, and information the coaching staff has but the model does not.

The point estimates tell us the size of the edge. If going for it is estimated at 0.559 win probability and kicking a field goal is estimated at 0.541, the estimated edge is 0.018. But the magnitude of that difference is not itself a measure of stability. To assess stability, we need an estimate of the variability in the fitted pipeline.

One way to estimate that variability is to refit the whole pipeline on bootstrap samples. For each refit, recompute the three action values for the same game state and record which action is best. If going for it wins almost every refit, the recommendation is more stable. If going for it and kicking trade places often, the recommendation is less stable, regardless of the original point estimate.

With B bootstrap refits, one useful summary is

$$\hat{\mathbb{P}}(\text{go is best}) = \frac{1}{B} \sum_{b=1}^B \mathbb{1} \left\{ \widehat{WP}_{\text{go}}^{*(b)}(x) > \max(\widehat{WP}_{\text{FG}}^{*(b)}(x), \widehat{WP}_{\text{punt}}^{*(b)}(x)) \right\}.$$

This quantity is the fraction of bootstrap refits in which going for it is the estimated best action. Values near 1 indicate stable support for going; lower values indicate that refitting often favors another action. The same refits can also be used to track how much the estimated action gap moves under refitting. That does not replace the point estimate. It tells us how sensitive the estimated edge is to perturbations in the fitted pipeline. For football play by play data, the resampling unit matters. Resampling individual plays can be too optimistic when plays from the same game share unmodeled context.

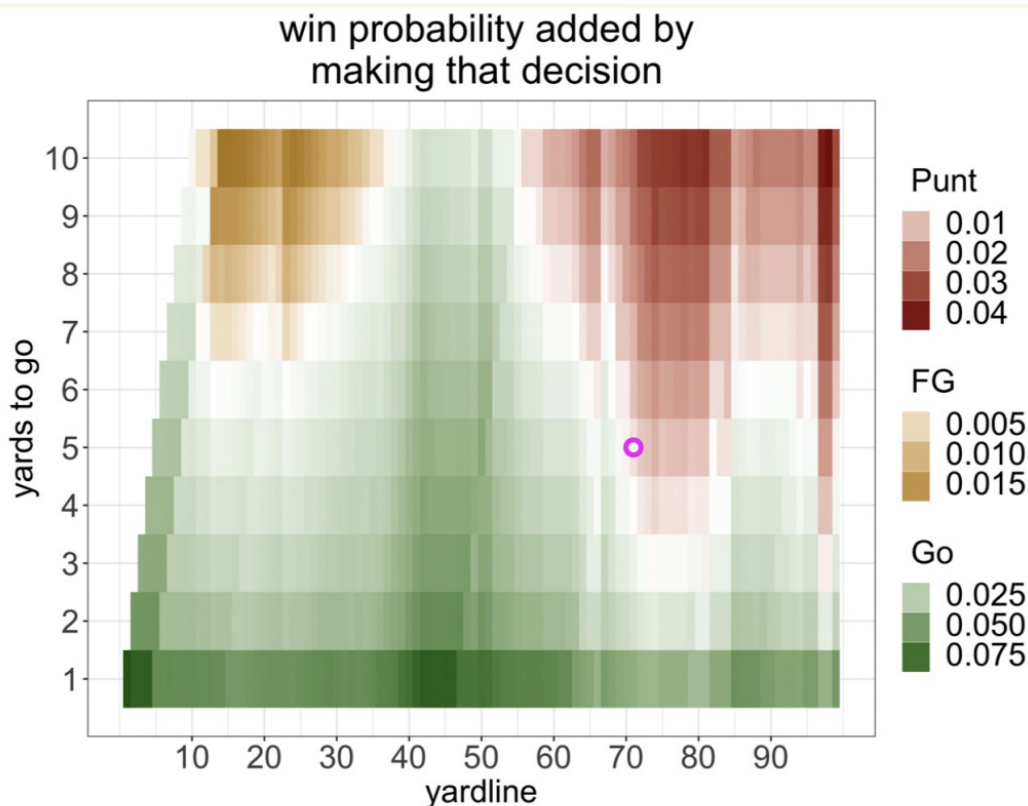


Figure 20.2: Estimated win probability gain from Brill's fourth down decision model. The point estimate gives the magnitude of the edge; refit variability checks how stable that edge is.

20.2.2 Reporting Fourth Down Recommendations

The reported recommendation should match both the estimated edge and the stability check. Table 20.1 gives one possible vocabulary.

Label	Pattern	Suggested wording
Clear	Positive edge; stable across refits	The fitted win probability model favors going for it.
Lean	Positive edge; mixed refit support	The point estimate favors going for it, but the call is sensitive.
Too close	Small edge or unstable best action	The model treats the options as essentially tied.
Not covered	Missing context or outside training support	This case is outside the model's checked support.

Table 20.1: Language for fourth down recommendations.

For fourth down decisions, the reporting task is to name the objective, the size of the estimated edge, and how stable the recommendation is under refitting. In that sense, the last column is part of the analysis, not an after the fact caveat. Rankings create the same problem in a different form: an ordered list can look more certain, and more general, than the fitted model justifies.

20.3 Case Study 2: Rankings From Derived Metrics

In Lecture 14, a Bayesian power rating model estimated team strengths, measured as expected point differential relative to league average. The model did not just return one ordered list. Posterior draws gave plausible team strength values after seeing the data. Within one draw, teams can be ranked. Across draws, the order can change. The reporting problem begins when that uncertainty is compressed into a single leaderboard.

If θ_j is team j 's strength, a point ranking orders $\hat{\theta}_j$. But θ_j is not generic team quality. In Lecture 14, it was a latent score margin effect in a model with team indicators, home field advantage, and partial pooling. Change the response, the predictors, the season window, or the prior, and the ranking can change because the definition of strength and the evidence behind it changed. A posterior ranking can instead report

$$\mathbb{P}(\text{rank}_j \leq 5 \mid \text{data}) \approx \frac{1}{M} \sum_{m=1}^M \mathbb{1}\{\text{rank}(\theta_j^{(m)}) \leq 5\}.$$

A pairwise comparison can report

$$\mathbb{P}(\theta_A > \theta_B \mid \text{data}).$$

These are different claims:

- Team A has the third highest posterior mean.
- Team A is top five in 82% of posterior draws.
- Team A is stronger than Team B in 63% of posterior draws.

The first sentence reports an ordered point estimate. The second reports rank uncertainty. The third reports pairwise uncertainty. They are different summaries of the same posterior, and they should not be confused with a general ranking of team quality.

20.3.1 Leaderboards Select Extremes

Uncertainty is only part of the issue. A leaderboard also depends on what quantity the model was built to estimate. Large observed or fitted values are often a mix of signal and favorable noise. That is why shrinkage appeared in several parts of the course:

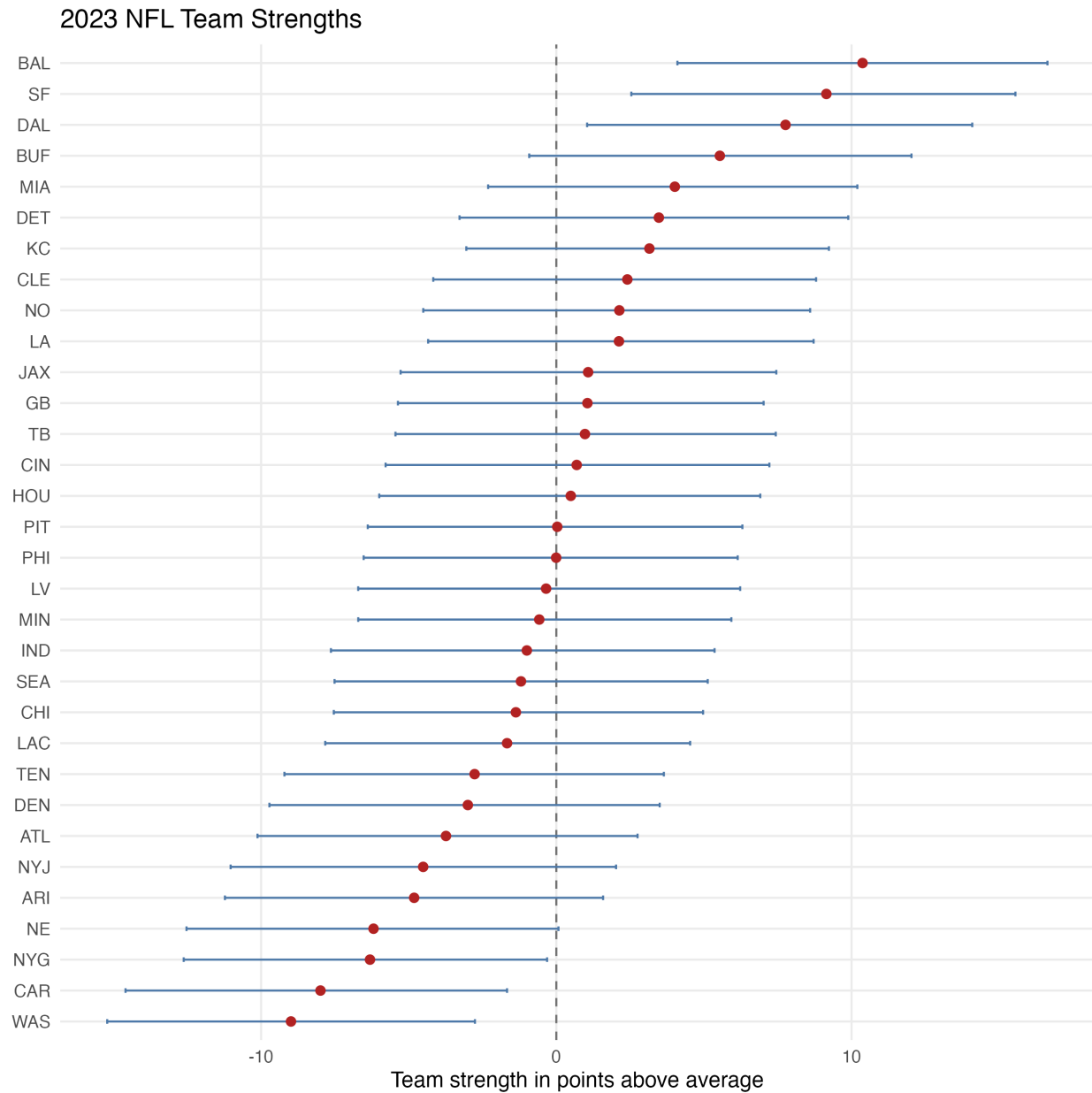


Figure 20.3: Posterior means and 95% credible intervals for team strength from Lecture 14. Exact rankings are less stable when intervals overlap.

- empirical Bayes moved noisy batting averages toward a league learned prior;
- shrinkage estimation reduced joint prediction risk;
- ridge regression stabilized adjusted plus minus;
- hierarchical Bayesian models pooled related team effects.

Player impact metrics make this concrete. Raw plus minus, adjusted plus minus, and regularized adjusted plus minus are different ways to turn on court scoring data into a player ranking. In RAPM, the response is possession margin, the predictors are lineup indicators, and regularization penalizes large coefficients. The resulting estimate is about lineup adjusted point impact in that sample, not total player value.

Three Plus-Minus Estimates

Same 2023-24 possession data; estimates shown in points per 100 possessions

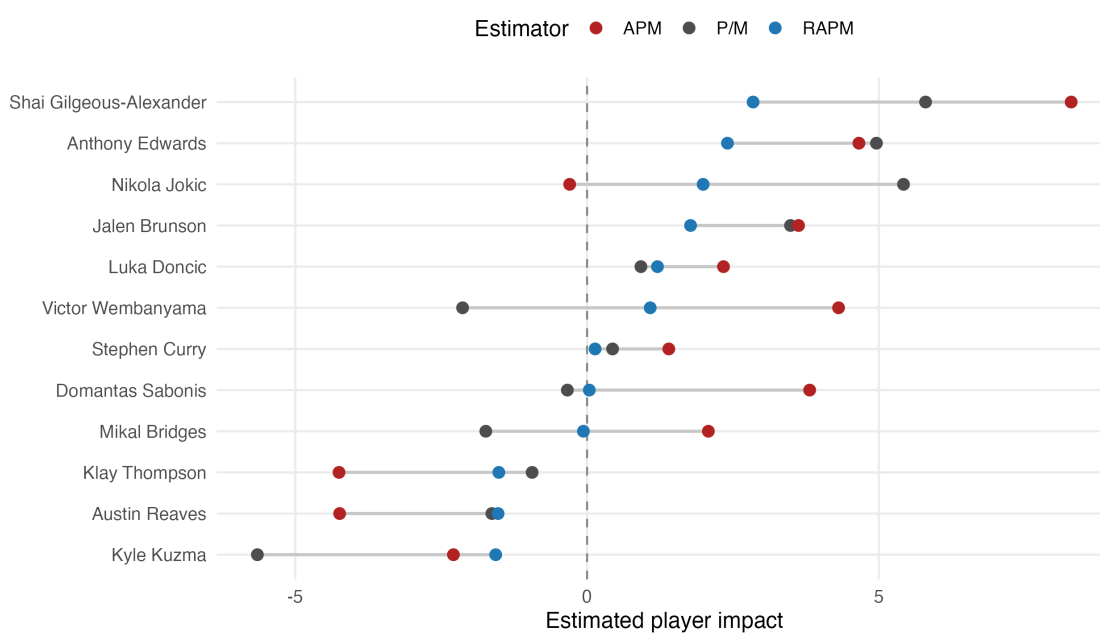


Figure 20.4: Raw plus minus, adjusted plus minus, and regularized adjusted plus minus from Lecture 15. Regularization changes the objective by penalizing large player coefficients.

The RAPM objective is roughly

fit possession margins + penalize large player coefficients.

That is a useful objective for estimating lineup adjusted point impact. Change the response, the design matrix, or the penalty, and the reported metric changes with it. It does not capture health, contract value, aging, playoff fit, role portability, or tactical responsibilities that are absent from the design matrix.

A narrower claim supported by the model is: “In this sample, under this lineup adjusted and regularized model, this player has the higher estimated impact.” If nearby estimates overlap, the comparison should say that too. Rankings overstate conclusions in two ways: they hide uncertainty, and they can hide what quantity was ranked in the first place.

20.3.2 Reporting Rankings

Overstated claim	Model supported claim
This is the eighth best player.	This player has the eighth highest point estimate, with uncertainty that overlaps several nearby players.
Team A is better than Team B.	The model estimates Team A higher; in the posterior draws, Team A is stronger than Team B 68% of the time.
The metric proves Player X is elite.	The metric is evidence about Player X's estimated lineup adjusted point impact in this sample and model. It should be interpreted alongside role and context.
The ranking is objective.	The ranking follows from choices about response, predictors, target, shrinkage, and validation.

Table 20.2: Language for rankings and derived player metrics.

20.4 From Estimates to Claims

The two examples use different models, but the reporting problem is the same. Before presenting a result, check that the reported claim answers the questions in Table 20.3.

Question	What to report
Target	What quantity is being estimated: a mean, probability, rank, causal effect, or decision value?
Data	Which seasons, leagues, games, players, or plays were used? What population is the analysis meant to describe?
Variables	What outcome was modeled? What information did the model see? What important information was left out?
Objective	What did the fitting procedure optimize: squared error, log loss, likelihood, posterior probability, a penalty, or expected utility?
Uncertainty	What changes if we resample, refit, use posterior draws, change priors, or perturb reasonable modeling choices?
Validation	Was the model checked on held out data, future seasons, calibration, posterior predictive checks, or stability tests?
Decision	If an action is recommended, what objective turns the estimate into that action?

Table 20.3: A short checklist for moving from a model output to a statistical claim.

For a decision problem, name the objective. For a ranking or derived metric, report uncertainty around the order and name the quantity being ranked, including the outcome and predictors used to define it.

20.5 Takeaways

- Statistical honesty means matching the claim to the target, data, model, and uncertainty.
- A decision recommendation is conditional on an objective function.
- A small estimated edge should be reported as small, even when it points in one direction.

- Rankings should report uncertainty about ranks and name the quantity being ranked.
- A derived metric is defined by its response, predictors, and fitting objective; it is not a direct measurement of total player quality.
- A limitation matters when it changes how the estimate should be used.

References

- [BYW] Brill, R. S., Yurko, R., & Wyner, A. J., *Analytics, Have Some Humility: A Statistical View of Fourth-Down Decision Making*, The American Statistician, 2025.
- [BR] Brill, R. S., *Fourth-down decision model and Shiny application*, GitHub repository.
- [BB] Baldwin, B., *NFL Win Probability from Scratch using XGBoost in R*, April 16, 2021.
- [ET] Efron, B., & Tibshirani, R. J., *An Introduction to the Bootstrap*, Chapman and Hall, 1993.
- [BDA] Gelman, A., Carlin, J. B., Stern, H. S., Dunson, D. B., Vehtari, A., & Rubin, D. B., *Bayesian Data Analysis*, CRC Press, 2013.